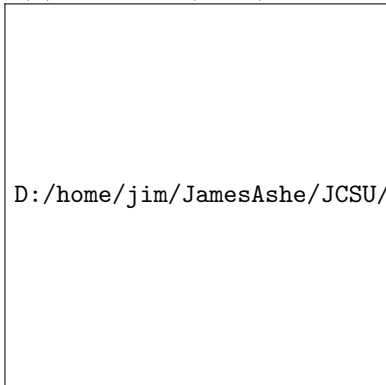


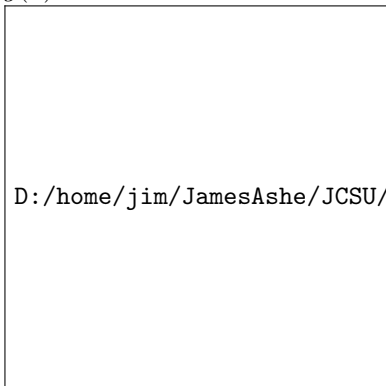
Sketch a graph of the functions, give the equation of the parabola's axis of symmetry, and determine the domain and range.

1. $f(x) = y - 3 = (x - 1)^2$



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2. $g(x) = x^2 - 2x - 15$



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Solve the problem.

3. A farmer has 100 feet of fence, and she wants to make a rectangular pen. What length and width should she use if she wants to enclose the maximum amount of area?